

Roberto Olvera-Hernandez

Bioinformatics Software Engineer

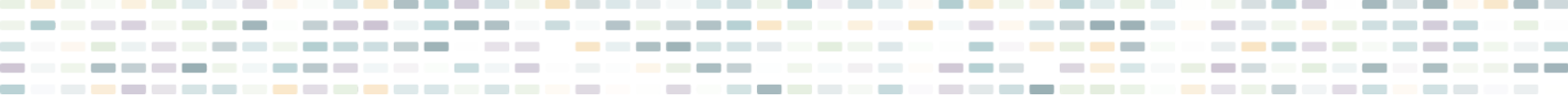
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Professional & Research Experience

Feb. 2025 - Present	Computational Genomics Researcher Center for Genomic Sciences (CCG) – UNAM Cuernavaca, MX Leading the design and implementation of scalable bioinformatics pipelines for processing genomic data from over 100,000 individuals on cloud infrastructure (DNAnexus; AWS/Azure). • Software Development: Engineered an ETL pipeline for large-scale genome-wide association studies (GWAS), optimizing data transformation and variant-calling workflows for high-throughput analyses. • AI-Assisted Engineering: Leveraging AI coding tools for rapid prototyping, pipeline testing, and automation of complex data workflows. • Cross-functional Collaboration: Partnering with international teams to ensure production-readiness and robust data governance for large-scale genetic pipelines.
Aug. 2025 - Jan. 2026	Genetic Epidemiology — Research Placement Nuffield Department of Public Health (NDPH) – University of Oxford Oxford, U.K. Developed and maintained a robust computational pipeline for within-family GWAS, directly addressing confounding from population structure in highly admixed cohorts. • Data Pipelines: Architected and deployed cloud-native pipelines for epidemiological data analysis, transitioning complex statistical models into scalable workflows. • Statistical Genetics: Implemented advanced statistical genetics tools to extract unbiased heritability and genetic correlation estimates.
2024 - 2025	Income Data Analyst TOKA Internacional Zapopan, MX Spearheaded the modernization of a fragile billing process for millions of monthly transactions, navigating a rapid migration from local servers to Azure. • ETL & Orchestration: Architected and deployed an automated ETL pipeline using Python and Dagster, replacing fragile manual SQL processes and ensuring reliable fee processing. • Business Intelligence: Automated the data delivery lifecycle, integrating the pipeline to publish results directly to Power BI and auto-email PDF/Excel reports to collaborators. • Cloud Migration: Successfully adapted the data architecture and orchestration logic to Azure during a rapid cloud transition.
2021 - 2022	Diagnostic Data Trainee Boehringer Ingelheim – Vetmedica Zapopan, MX Bridged wet-lab operations and bioinformatics within an animal health laboratory, supporting viral genomic surveillance for veterinary vaccine development. • AI Data Processing: Engineered an automated data pipeline using AI tools to extract veterinary diagnostic information from unstructured PDFs directly into a relational database. • Lab Systems & NGS: Executed end-to-end wet-lab protocols including Illumina sequencing and qPCR, building a strong understanding of sample processing and DNA/RNA analysis. • Bioinformatics: Conducted preliminary viral evolution tracking across agricultural sites, analyzing genomic data to support animal health and vaccine research efforts.

Education

2025 - Present	Ph.D. in Biomedical Sciences (Genomics Focus) Center for Genomic Sciences (CCG) – UNAM Cuernavaca, MX
2018 - 2023	B.Sc. in Biotechnology Engineering ITESO Guadalajara, MX



⚙ Technical Skills

Programming: Python, R, Bash, SQL, Git
Cloud/Infra: DNAnexus, AWS (basics), Linux
Genomics: Population & Statistical Genetics, GWAS
AI Tools: Agentic coding tools, Prompt engineering

Languages

Spanish: Native
English: Fluent (TOEFL iBT - C1)

⚙ Certifications

Jul. 2023	Bioinformatic Tools for High-Throughput Sequencing USMBB - IBt UNAM
Aug. 2020	Data Science: R Basics EBRC